

30

Formula of Macro

$$\text{Net} = \text{Gross} - \text{Depreciation}$$

$$\text{Gross} = \text{Net} + \text{Depreciation}$$

$$\text{Market Price} = \text{Factor Cost} + \text{NIT (Net Indirect Tax)}$$

$$\text{Factor Cost} = \text{Market Price} - \text{NIT}$$

$$\text{NIT} = \text{Indirect Tax} - \text{Subsidies}$$

$$\text{National Income} = \text{Domestic Income} + \text{NFIA}$$

$$\text{NFIA} = \text{FIFA} - \text{FITA}$$

→ By Value Added Method (GVA_{MP} or GDP_{MP})

$$\text{GVA}_{\text{MP}} \text{ or } \text{GDP}_{\text{MP}} = \text{value of output} - \text{Intermediate consumption}$$

$$\text{Value of Output} = \text{Sales} + \text{change in stock}$$

$$\text{change in stock} = \text{Closing Stock} - \text{Opening Stock}$$

$$\text{Sales} = \text{Quantity} \times \text{Price}$$

→ By Income Method (NDP_{FC})

$$\text{NDP}_{\text{FC}} = \text{compensation of Employees} + \text{operating Surplus} + \text{Mixed Income of Self employed}$$

Compensation of Employees = wages and salaries in cash + wages and salaries in kind + social security schemes by employer

$$\text{Operating Surplus} = \text{Rent and Royalty} + \text{Interest} + \text{Profit}$$

$$\text{Profit} = \text{Corporate Tax} + \text{Undistributed Profit} + \text{Dividend}$$

→ By Expenditure Method (GDP_{Mp})

$$\text{GDP}_{Mp} = \text{Private final Consumption Expenditure} + \text{Govt. final Consumption Expenditure} + \text{Gross Domestic Capital formation} + \text{Net Exports} - \text{Imports}$$

Gross Domestic Capital formation = Gross Domestic fixed Capital formation + Change in stock

Nominal GDP = current Quantity x current Price

Real GDP = Base year Quantity x Base year Price

$$\frac{\text{Real GDP or GDP at constant Price}}{\text{GDP at current Price}} = \frac{\text{GDP at current Prices}}{\text{Price Index}} \times 100$$

$$\frac{\text{Nominal GDP or GDP at current Price}}{\text{GDP at current Price}} = \text{Real GDP} \times \frac{\text{Price Index}}{100}$$

Income and Employment

- Aggregate Demand (AD) = consumption (C) + Investment (I)
- Aggregate Supply = Income (Y)
- Income (Y) = consumption (C) + saving (S)
- $APC = \frac{C}{Y}$ (APC = Average Propensity to Consume)
- $APS = \frac{S}{Y}$ (APS = Average Propensity to Save)
- $MPC = \frac{\Delta C}{\Delta Y}$ (MPC = Marginal Propensity to consume)

- $MPS = \frac{\Delta S}{\Delta Y}$ [MPS - Marginal Propensity to Save]

- $APC + APS = 1$

- $MPC + MPS = 1$

- $C = \bar{C} + b(Y)$

- $S = \bar{S} + b(Y)$ or $S = -C + (1-b)Y$

- Break Even Point = $S=0$ or $C=Y$

Investment Multiplier -

$$K = \frac{1}{1-MPC} \quad \text{or} \quad K = \frac{1}{MPS}$$

$$K = \frac{\Delta Y}{\Delta I}$$

Government Budget

- Revenue Deficit = Revenue Expenditure - Revenue Receipt
- Fiscal Deficit = Total Expenditure - Total Receipts (excluding borrowings)
- Primary Deficit = Fiscal deficit - Interest Payments

Balance of Payment

- Balance of Trade = Export of goods - Import of goods
- Balance of Payment = Inflow of foreign exchange - outflow of foreign exchange